

MMML



end-to-end MLIP toolkit

// data · train · MD · spectra · deploy · 31+ CLI
commands

\$ mmml -help

// COSMO Lab · EPFL

 [EricBoittier/mmml](#)

 [docs](#)

 [EricBoittier/mmml_tutorial](#)

 [CI](#)

// contributions from Sena Aydin, Khamlek Chaton, and
everyone who has built MMML

MLIP workflow — generate · train · simulate · analyze

GPU reference data

```
mmml pyscf-dft
mmml pyscf-mp2
mmml pyscf-evaluate
mmml normal-mode-sample
mmml xml2npz
mmml fix-and-split
mmml validate
mmml active-learning
```

train & evaluate

```
mmml physnet-train
mmml train-joint
mmml ef-train
mmml physnet-evaluate
```

MD & hybrid MM/ML



```
mmml md-system
mmml physnet-md
mmml ef-md
```

// CHARMM MLPot · JAX-MD · λ -dynamics TI

spectra & conformers

```
mmml-spectra-md
// IR · VCD · Raman
mmml orca-server
// Ext0pt GOAT
mmml gui
```

```
mmml lambda-mbar
```

Hybrid CHARMM integration is one deployment path — not the whole repo.

Atomistic ML software — MMML today, Metatomic next?

mmml — zsh

CI/CD · tests

✓ python -m build pass

✓ pytest tests/ pass

✓ mkdocs build --strict pass

65+ unit tests · pycharmm | gpu | mlpot

// Python 3.13 · uv · JAX JIT · Pydantic · NPZ schema ·
Ruff · mypy

mmml — zsh

// metatensor/metatomic · ORCA ExtOpt



issue 228



PR metatensor/
metatomic#262 +1328 ●
open

PR 262: metatomic-orca-server / client

MMML: mmml orca-server · GPU batch · FastAPI

Same ExtOpt protocol in JAX (MMML) and metatomic-torch →
COSMO